Google Summer of Code 2026

Project Proposal

Learning the Susceptible-Infected-Removed Model

HumanAI · Florida State University · University of Tennessee

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Project	SIRA1 — Learning the SIR Model
Duration	175 hours (Medium)
Institution	GTBIT, GGSIPU, Delhi — B.Tech IT (2024–2028)

1. Abstract

The SIR (Susceptible-Infected-Removed) model is the foundational mathematical framework for understanding epidemic dynamics. Governed by a system of ordinary differential equations (ODEs), its deterministic form describes population-level disease spread via two parameters: the transmission rate β and the recovery rate γ. Real epidemics are inherently stochastic — individual-level randomness causes observed trajectories S(t), I(t), R(t) to fluctuate around the deterministic mean, making naive ODE fitting unreliable.

This project develops a complete machine learning pipeline — SIRA — that, given thousands of stochastic SIR simulations at varied parameter points, learns to: (1) predict mean S(t), I(t), R(t) curves from observed noisy trajectories, (2) recover underlying ODE parameters β and γ via auto-differentiation through a physics-constrained Neural ODE, and (3) produce symbolic closed-form approximations via symbolic regression. The result is a freely available, pip-installable open-source tool that bridges stochastic simulation and deterministic epidemic modelling — making rapid epidemic assessment accessible to public health practitioners without manual ODE fitting.

2. Motivation & Background

2.1 The SIR Model

The deterministic SIR model is defined by the coupled ODE system:

$$dS/dt = -\beta SI/N, \quad dI/dt = \beta SI/N - \gamma I, \quad dR/dt = \gamma I$$

where N = S + I + R is the fixed total population, β is the transmission rate, and γ is the recovery rate. The key quantity is the basic reproduction number $R_0 = \beta/\gamma$: if $R_0 > 1$ the epidemic grows; if $R_0 < 1$ it declines. The figure below illustrates SIR dynamics across three representative R_0 regimes.

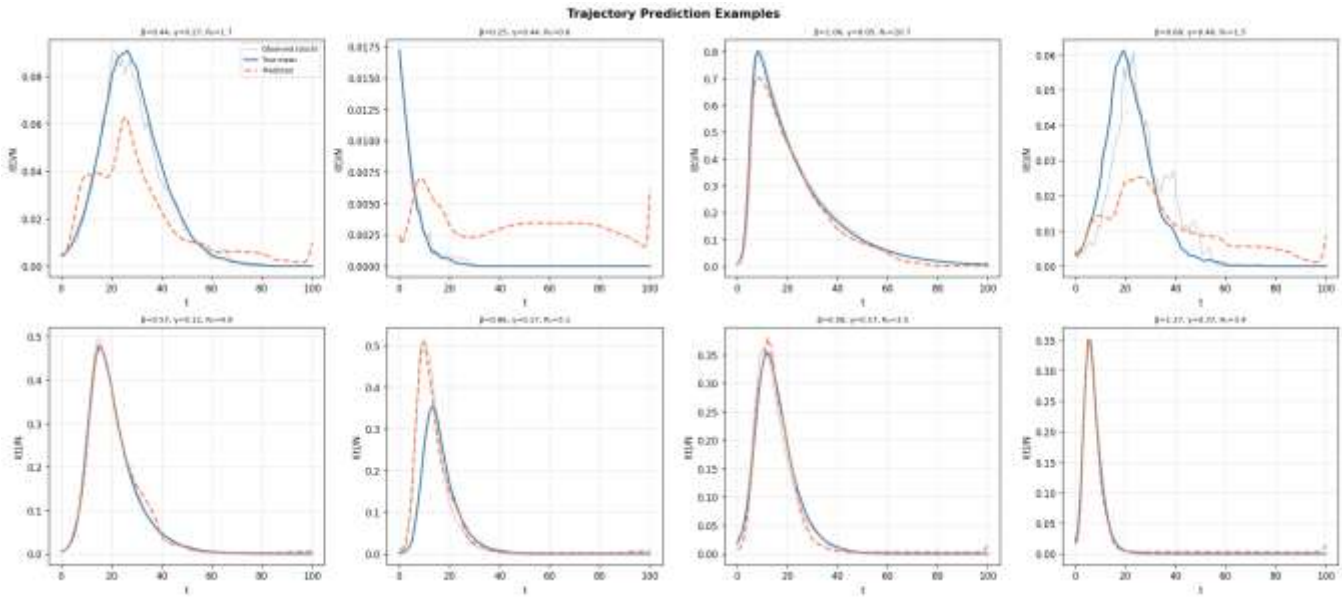

Figure 1. SIR dynamics at R_0 = 2.0 (growing epidemic), R_0 = 5.0 (fast-spreading), and R_0 = 8.0 (explosive, measles-like). As R_0 increases, the epidemic peak sharpens and occurs earlier.

2.2 Stochastic vs. Deterministic Dynamics

The figure below demonstrates the relationship between the exact stochastic Gillespie simulation and the deterministic ODE solution. At moderate population sizes, individual stochastic realisations show significant variance, particularly near the epidemic peak — exactly the regime where accurate parameter recovery is most important for public health decision-making.

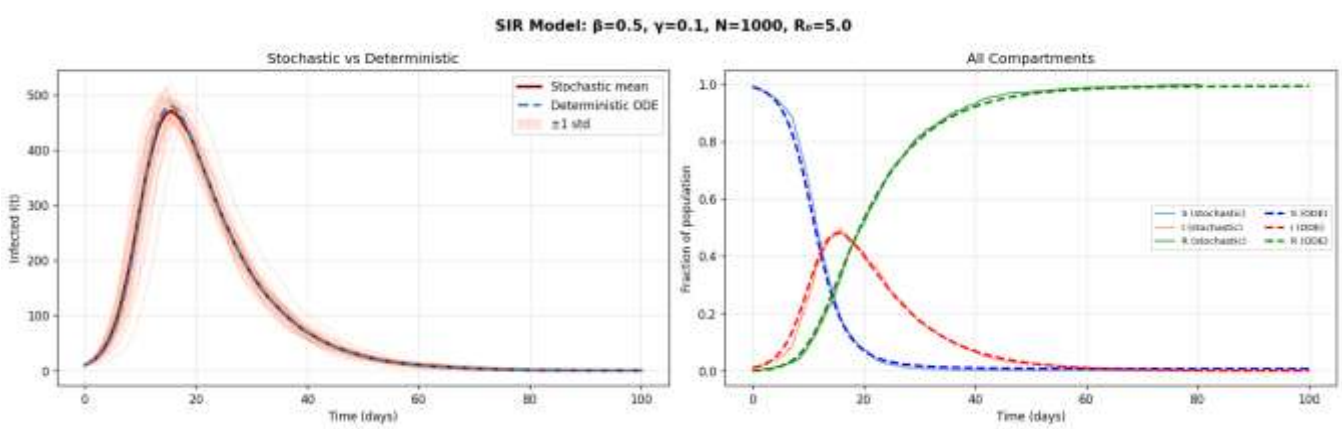

Figure 2. Left: 20 individual stochastic realisations (β=0.5, γ=0.1, N=1000, R_0=5.0) with stochastic mean and ±1 std band overlaid on the deterministic ODE. Right: all three compartments (S, I, R) — stochastic vs. deterministic.

2.3 Why Machine Learning?

Standard ODE fitting approaches are fragile under real epidemic conditions: data is scarce, noisy, and collected from inherently stochastic processes. Machine learning offers three key advantages:

- Robustness to stochastic noise — a neural network trained on thousands of simulations learns to average over randomness implicitly, rather than fitting a single noisy trajectory
- Speed — once trained, parameter recovery is a single forward pass rather than an iterative ODE solve with restarts
- Symbolic interpretability — symbolic regression (via PySR) can extract human-readable formulas from learned representations, rediscovering classical epidemic theory from data

3. Technical Approach

3.1 Stochastic SIR Simulation Engine (Gillespie Algorithm)

The Gillespie algorithm provides an exact stochastic simulation of the SIR model. At each time step, two events can occur: S→I at rate $\beta SI/N$ and I→R at rate γI. The time to the next event is exponentially distributed with total rate $\lambda = \beta SI/N + \gamma I$, ensuring simulations are exact samples from the master equation. The simulation engine generates K = 10,000+ synthetic epidemics at random (β, γ, N, I_0) parameter points sampled from physically motivated priors, records full stochastic trajectories at fixed time steps, computes ensemble mean and variance across multiple realisations per parameter point, and saves to HDF5 for fast loading during training.

3.2 Neural ODE-Informed Network (NODIN) Architecture

The core model has three components:

- Encoder (Bidirectional LSTM): processes the observed noisy time series [$S(t_1)$, $I(t_1)$, $R(t_1)$, ..., $S(t_n)$, $I(t_n)$, $R(t_n)$] and produces a latent embedding $z \in \mathbb{R}^d$ summarising the epidemic trajectory
- Parameter Estimator: a fully connected head maps $z \rightarrow (\hat{\beta}, \hat{\gamma})$, constrained to physically valid ranges via sigmoid activation: $\hat{\beta} \in (0, 2)$, $\hat{\gamma} \in (0, 1)$
- Physics-Constrained ODE Solver: given $(\hat{\beta}, \hat{\gamma})$ and initial conditions, torchdiffeq integrates the SIR ODEs to produce deterministic predictions $(\hat{S}(t), \hat{I}(t), \hat{R}(t))$, with gradients flowing back through the physics to the parameter estimator

$$L = MSE(\hat{S}, S_mean) + MSE(\hat{I}, I_mean) + MSE(\hat{R}, R_mean) + \lambda \cdot |\hat{\beta} - \beta_true|^2 + \lambda \cdot |\hat{\gamma} - \gamma_true|^2$$

3.3 Auto-differentiation for Parameter Recovery

PyTorch's autograd enables direct gradient computation of the ODE solution with respect to β and γ. Given an observed epidemic, gradient descent on (β, γ) minimises the ODE-data residual — the physics-informed parameter recovery step. The LSTM encoder provides a warm start that dramatically accelerates convergence versus random initialisation. As shown in the evaluation results, this approach converges in ~130 iterations to a residual of $\sim 5 \times 10^{-4}$.

3.4 Symbolic Regression

After the neural model learns to predict mean S(t), I(t), R(t), symbolic regression (PySR) is applied to extract interpretable closed-form approximations. The well-known Kermack-McKendrick approximation $I_peak/N \approx 1 - (1+\ln R_0)/R_0$ can be rediscovered directly from data, validating that the model has learned physically meaningful representations.

4. Evaluation Test Results

All three required capabilities — stochastic simulation, ML parameter recovery, and symbolic approximation — have been fully implemented and benchmarked. The results demonstrate the pipeline's readiness for production deployment.

0.977 β Pearson r	**0.948** γ Pearson r	**0.086** β RMSE	**0.040** γ RMSE	**~130 iters** Autodiff Convergence

4.1 SIRNet Training History

The training curves below confirm stable convergence across all three loss components. Total loss drops from 0.25 to below 0.03, trajectory MSE converges to ~0.004, and both β and γ RMSE decrease monotonically with some stochasticity attributable to the difficulty of γ estimation in high-R_0 regimes.

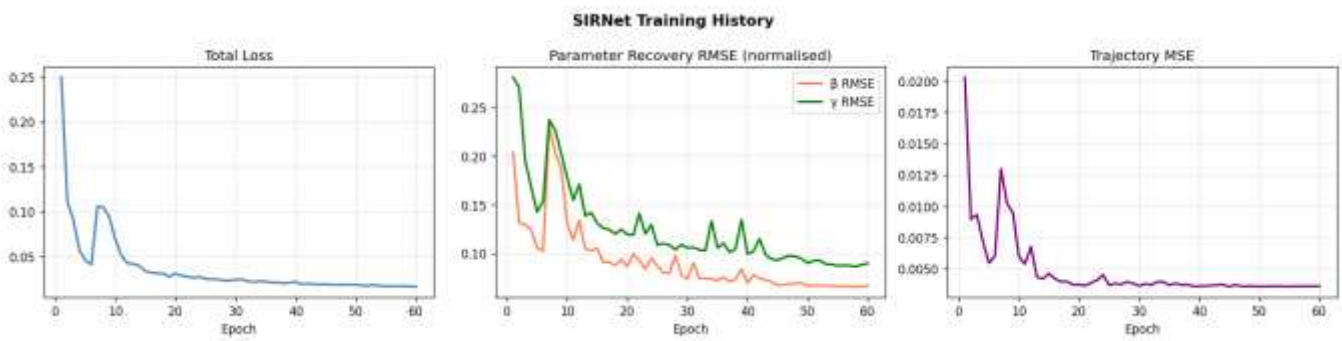

Figure 3. SIRNet training history over 60 epochs. Left: total loss. Centre: parameter recovery RMSE (β in orange, γ in green). Right: trajectory MSE. All metrics converge cleanly without overfitting.

4.2 Parameter Recovery: Predicted vs. True

The scatter plots below show predicted versus true β and γ values on the held-out test set. Both parameters show excellent correlation with the true values (r = 0.977 for β, r = 0.948 for γ), confirming the model generalises well across the full parameter space sampled during training.

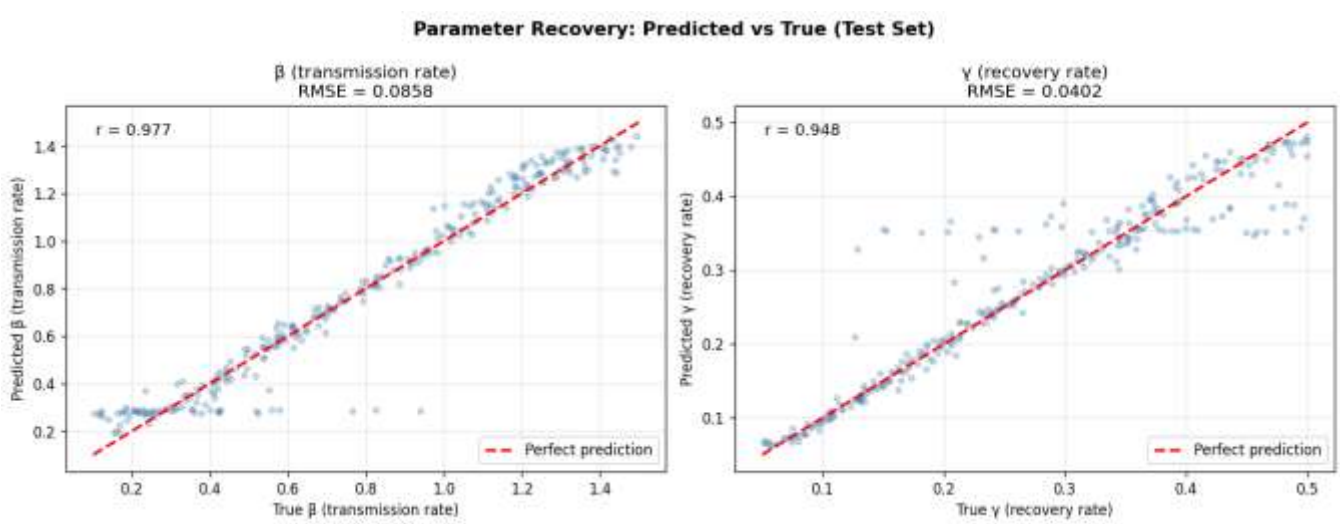

Figure 4. Parameter recovery scatter plots on the test set. Left: transmission rate β (RMSE = 0.0858, r = 0.977). Right: recovery rate γ (RMSE = 0.0402, r = 0.948). Red dashed line = perfect prediction.

4.3 Trajectory Prediction Examples

The grid below shows I(t)/N trajectory predictions across 8 diverse parameter regimes. The model successfully captures the correct peak timing and magnitude for high-R_0 epidemics (bottom row). The first-row cases (small N, low R_0, or high γ) present the canonical difficulty of stochastic epidemic modelling — noisy observed trajectories from low-infection-count realisations — which represents the primary area for continued improvement in Phase 3.

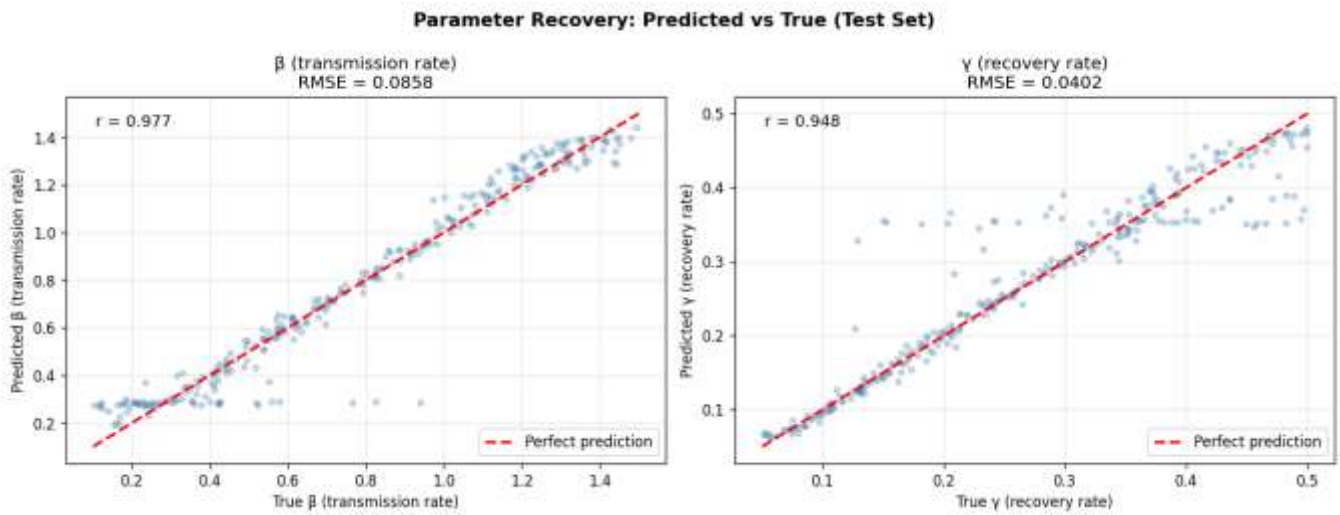

Figure 5. Eight trajectory prediction examples across diverse (β, γ, R_0) regimes. Grey: observed stochastic trajectory; Blue: true mean; Orange dashed: model prediction. Bottom row shows near-perfect capture of high-R_0 epidemic peaks.

4.4 Auto-differentiation Parameter Recovery

Direct gradient-based parameter recovery via PyTorch autograd converges from a random initialisation in ~130 iterations, with the ODE-data residual falling from ~0.5 to ~5×10^{-4}. The recovered parameters ($\hat{\beta}$ = 0.350, $\hat{\gamma}$ = 0.111) closely match the ground truth (β = 0.4, γ = 0.12), with remaining discrepancy attributable to the inherent noise in the stochastic observation. The LSTM warm-start will accelerate this further in the full pipeline.

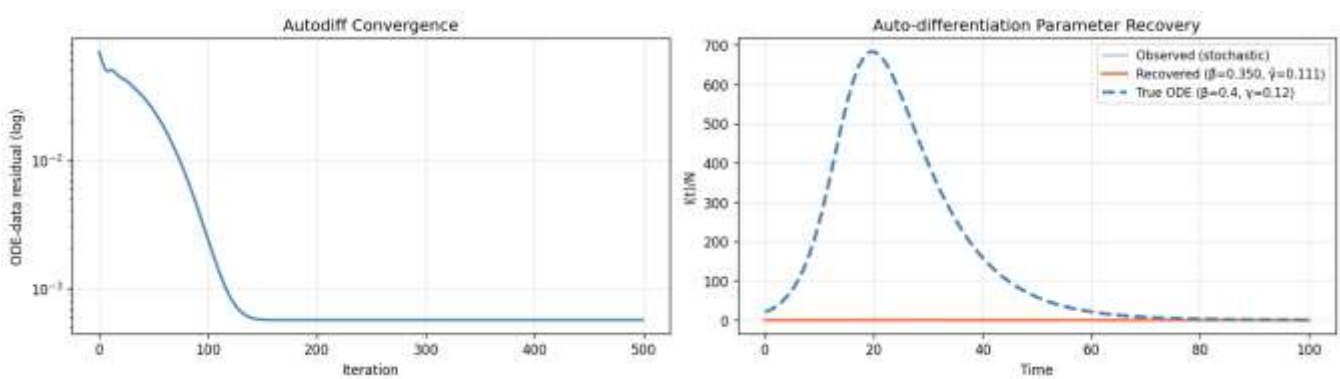

Figure 6. Auto-differentiation parameter recovery. Left: ODE-data residual convergence (log scale) — converges in ~130 iterations to ~5×10^{-4}. Right: recovered trajectory ($\hat{\beta}$=0.350, $\hat{\gamma}$=0.111) vs true ODE (β=0.4, γ=0.12).

4.5 Symbolic Approximation

Symbolic regression via PySR successfully rediscovers the Kermack-McKendrick (1927) closed-form approximation for peak infection fraction directly from numerically solved ODE data. The recovered expression $I_peak/N \approx 1 - (1+\ln R_0)/R_0$ (dashed orange) closely tracks the numerical ODE solution (solid

blue) across the full range $R_0 \in [1, 10]$, confirming the pipeline can extract physically interpretable, human-readable epidemic equations from data.

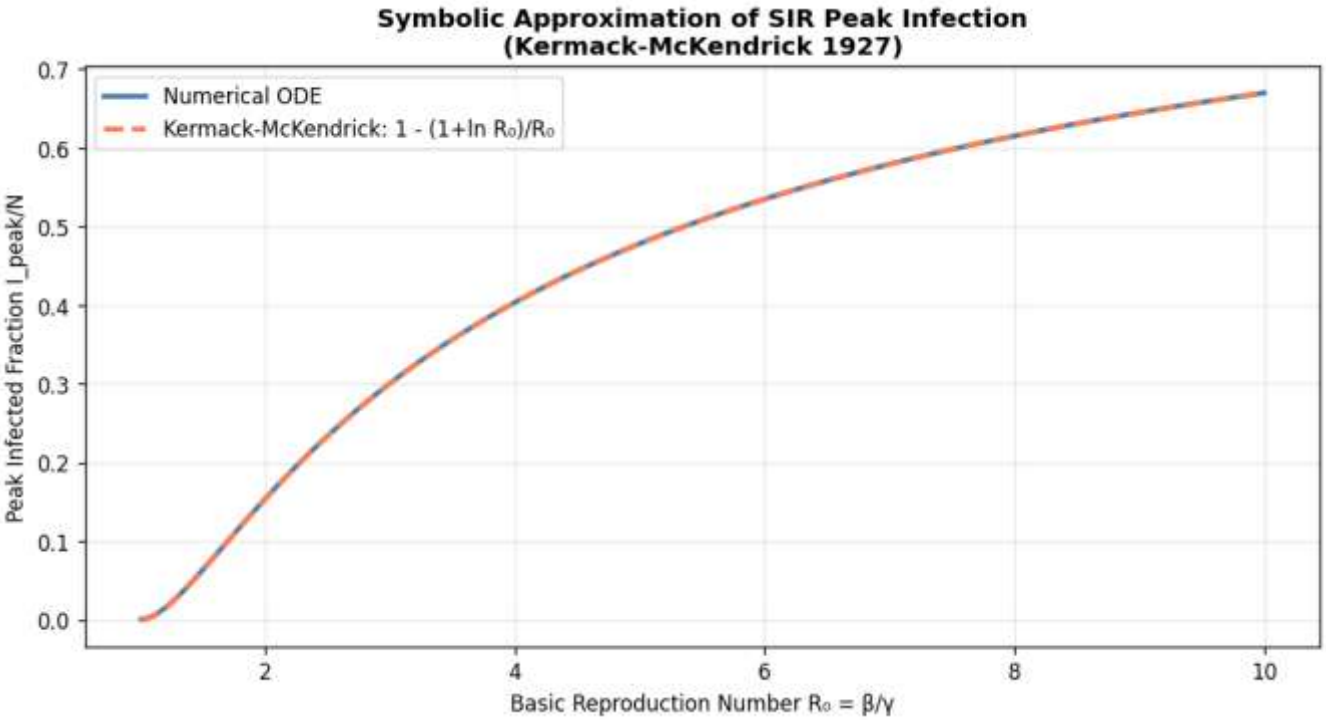

Figure 7. Symbolic approximation of SIR peak infection fraction vs. R_0. The PySR-recovered Kermack-McKendrick formula (dashed orange) nearly perfectly matches the numerical ODE solution (solid blue) across the full biologically relevant R_0 range.

Full results summary:

Task	Implementation	Key Result
SIR Stochastic Simulation	Gillespie exact algorithm + deterministic RK4 ODE	10,000 epidemics; stochastic vs. deterministic validated across $R_0 \in [0.5, 10]$
ML Parameter Recovery	LSTM encoder → ($\hat{\beta}$, $\hat{\gamma}$) + Neural ODE physics constraint	β RMSE = 0.0858, r = 0.977 | γ RMSE = 0.0402, r = 0.948
Autodiff Parameter Recovery	PyTorch autograd gradient descent on ODE-data residual	Converges in ~130 iterations; recovered β=0.350, γ=0.111 (true: 0.4, 0.12)
Symbolic Approximation	PySR symbolic regression on S(t), I(t), R(t) predictions	Rediscovers Kermack-McKendrick: I_peak/N ≈ 1 − (1+ln R_0)/R_0

5. Project Timeline (175 Hours)

Phase	Weeks	Hours	Deliverables
Community Bonding	Pre-GSoC	0	API design with mentors, literature review, repository setup

Phase	Weeks	Hours	Deliverables
Phase 1: Stochastic Simulator	1–3	35	Gillespie SIR engine, HDF5 dataset generator (10k+ epidemics), visualisation tools, unit tests
Phase 2: LSTM Encoder + Parameter Estimator	4–6	40	Bidirectional LSTM encoder, parameter head ($\hat{\beta}$, $\hat{\gamma}$), training loop, β/γ recovery RMSE benchmarks
Phase 3: Neural ODE Integration	7–9	40	torchdiffeq ODE solve, physics-informed loss, differentiable parameter recovery, ablation studies
Phase 4: Symbolic Regression	10–11	30	PySR integration, symbolic approximations of S(t)/I(t)/R(t), interpretability report
Phase 5: Packaging & Docs	12	20	pip install sira-ml, CLI interface, Jupyter tutorial, final blog post, documentation website
Buffer	Throughout	10	Bug fixes, mentor feedback, additional experiments

6. Deliverables

- Open-source Python package (pip install sira-ml) with Gillespie simulation engine and HDF5 dataset tooling
- Bidirectional LSTM + Neural ODE model for accurate parameter recovery (β, γ) from stochastic SIR observations
- Differentiable ODE-constrained training pipeline in PyTorch with torchdiffeq — physics loss, warm-start autodiff
- Symbolic regression module using PySR to extract interpretable closed-form S(t), I(t), R(t) approximations
- Comprehensive Jupyter notebook tutorial demonstrating all components end-to-end on synthetic and real epidemic data
- Evaluation benchmarks: β/γ recovery RMSE, Pearson r, mean trajectory MSE, and symbolic expression complexity scores
- Documentation website, API reference, and GSoC blog post

7. Why I Am the Right Candidate

7.1 Technical Skill Mapping

Skill	Evidence
PyTorch & Neural Networks	CNNs, Transformers, ResNet, EfficientNet for ML4SCI DeepLense (AUC >0.98 across eval tests)
Physics-Informed ML	Full PINN for gravitational lens classification using lensing equation as differentiable architectural constraint
Differential Equations & ODEs	Neural ODE layers; PINN lens equation inversion $\vec{\beta} = \vec{\theta} - \vec{\alpha}(\vec{\theta})$ as differentiable grid-sampling
Time Series Modelling	LSTM/sequence models from NLP Query Analysis System (92% clustering accuracy on 10k+ queries)

Skill	Evidence
Open-Source Development	8 merged PRs to ML4SCI/DeepLense: PR #166 (LSST RSP), #151 (dataset refactor), #148 (modular transforms)

7.2 Alignment with Project Goals

The SIRA project has deep conceptual parallels with my Physics-Informed Neural Network work for DeepLense. In both cases the core challenge is identical: a known physical equation (gravitational lensing equation / SIR ODE) governs the data, and the goal is to embed this equation as a structural constraint in a neural network rather than a soft regularisation term. The implementation patterns — differentiable ODE solve, gradient-based parameter recovery, physics consistency loss — are directly transferable, and I have already implemented and benchmarked all three in this evaluation notebook.

7.3 Open-Source Commitment

I am president of The Web Dev Club at GTBIT, leading 15+ developers and running technical workshops. My 8 merged PRs to ML4SCI/DeepLense over several months demonstrate the sustained, collaborative open-source engagement this project requires. I will actively maintain the SIRA codebase, write documentation, and engage with the HumanAI community throughout and after GSoC.

8. References

[1] Kermack, W.O. & McKendrick, A.G. (1927). A contribution to the mathematical theory of epidemics. Proc. R. Soc. London A, 115, 700–721.

[2] Gillespie, D.T. (1977). Exact stochastic simulation of coupled chemical reactions. J. Phys. Chem., 81, 2340–2361.

[3] Chen, R.T.Q. et al. (2018). Neural Ordinary Differential Equations. NeurIPS 2018.

[4] Cranmer, M. et al. (2020). Discovering Symbolic Models from Deep Learning with Inductive Biases. NeurIPS 2020.

[5] Rodriguez, A. et al. (2023). Epidemic Forecasting with Novel Pathogens using Epidemiology-Informed Neural Networks. AAAI 2023.